

Patterna Tuning Cheatsheet

This cheatsheet summarizes what happens when common Patterna parameters are increased or decreased.

Use this guide when outputs look wrong and you need to adjust segmentation, masking, spiderweb sampling, or phenotype classification settings.

Colony mask tuning

Parameter	Increase value	Decrease value	Use when
THRESH_PCT	Makes the colony mask stricter. Less area is included. May exclude dim colony regions.	Makes the mask more permissive. More area is included. May include background.	Adjust when the colony mask is too small or too large.
MASK_MODE = "dapi"	Uses DAPI to define the colony area. Good when nuclei fill the micropattern clearly.	Not applicable.	Best default when DAPI clearly outlines the colony.
MASK_MODE = "composite"	Uses multiple channels to define the colony area. Can rescue weak DAPI masks.	Not applicable.	Use when DAPI is weak but marker channels outline the colony.
MASK_MODE = "dapi_sauvola"	Uses a more sensitive local DAPI threshold. Can detect uneven DAPI signal.	Not applicable.	Use when standard DAPI masking fails due to uneven illumination or weak signal.

Spiderweb sampling tuning

Parameter	Increase value	Decrease value	Use when
N_ANGLES	More angular sampling directions. Smoother and more detailed angle-radius maps. Larger	Fewer angular directions. Faster and smaller outputs, but less angular detail.	Increase for publication-quality spatial maps. Decrease for quick testing.

Parameter	Increase value	Decrease value	Use when
	files and slightly slower runtime.		
N_R	More radial sampling points from center to edge. Smoother radial profiles.	Fewer radial points. Faster and smaller outputs, but less radial detail.	Increase if radial profiles look too coarse.
SIGMA_R	More radial smoothing. Profiles look smoother but sharp boundaries may blur.	Less smoothing. Preserves sharper transitions but profiles may look noisier.	Increase for noisy profiles. Decrease if spatial transitions are being over-smoothed.

StarDist nuclei segmentation tuning

Parameter	Increase value	Decrease value	Use when
STARDIST_PROB_THRESH	Stricter detection. Fewer nuclei are detected. Reduces false positives but can under-segment.	More permissive detection. More nuclei are detected. Can recover missed nuclei but may add false positives.	Decrease if StarDist under-segments. Increase if it detects debris or noise.
STARDIST_NMS_THRESH	Allows more nearby/overlapping detections to remain. Can help recover crowded nuclei.	More aggressive suppression of overlapping detections. May reduce duplicate detections but can merge/remove close nuclei.	Tune together with STARDIST_PROB_THRESH when nuclei are crowded.
STARDIST_MIN_SIZE	Removes more small objects. Reduces debris but may remove real small nuclei.	Keeps smaller objects. Can recover small nuclei but may include specks.	Set to None to turn off minimum area filtering.
STARDIST_MAX_SIZE	Allows larger detected objects to remain. May keep clumps.	Removes large objects. Can remove clumps but may remove valid large nuclei.	Set to None to turn off maximum area filtering.
STARDIST_BBOX_PAD	Includes more area	Crops tighter around	Increase if edge

Parameter	Increase value	Decrease value	Use when
	around the colony crop. Safer for edge nuclei but slower.	the colony. Faster but may cut off edge nuclei if too low.	nuclei are missing.

Hybrid segmentation tuning

Parameter	Increase value	Decrease value	Use when
HYBRID_MIN_ACCEPTABLE_NUCLEI	More likely to trigger dense watershed fallback.	More likely to keep the StarDist result.	Increase if StarDist often under-segments dense colonies. Decrease if fallback is triggered unnecessarily.

Dense watershed fallback tuning

Parameter	Increase value	Decrease value	Use when
DENSE_WS_MIN_AREA	Removes more small objects. Reduces debris/noise.	Keeps smaller objects. May recover small nuclei but can add specks.	Increase if tiny false positives are counted. Decrease if small nuclei are missing.
DENSE_WS_MAX_AREA	Allows larger objects/clumps to remain.	Removes larger objects.	Decrease if clumps are being counted as single cells. Increase if valid large nuclei are removed.
DENSE_WS_GAUSS_SIGMA	More smoothing. Reduces noise but can merge close nuclei.	Less smoothing. Preserves detail but may be noisier.	Increase for noisy DAPI. Decrease if nuclei are merging.
DENSE_WS_BG_SIGMA	Corrects broader background gradients.	Corrects more local background variation.	Adjust when illumination is uneven.
DENSE_WS_MIN_DISTANCE	Fewer watershed seeds. Less splitting, more merging.	More watershed seeds. More splitting, possible over-segmentation.	Decrease if nuclei are merged. Increase if nuclei are over-split.
DENSE_WS_BLOCK_SIZE	Uses a larger local thresholding	Uses a smaller local window. More	Adjust if foreground detection is uneven.

Parameter	Increase value	Decrease value	Use when
DENSE_WS_C	window. Better for broad gradients. Changes adaptive threshold permissiveness. Direction can be image-dependent, so tune visually.	sensitive to local changes. Changes adaptive threshold permissiveness. Direction can be image-dependent, so tune visually.	Adjust only after checking QC overlays.

Cell-intensity map tuning

Parameter	Increase value	Decrease value	Use when
CELL_INTENSITY_NORM_PCTS_upper_percentile	Makes the display scale less sensitive to very bright outliers. May make dim signal easier to see.	Makes the scale include more of the full intensity range. Bright cells may dominate.	Tune when cell-intensity maps look saturated or too dim.
CELL_INTENSITY_NORM_MODE = "global_percentile"	Uses one display scale per marker across all images. Best for comparison.	Not applicable.	Use for comparing MPs in the same run.
CELL_INTENSITY_NORM_MODE = "per_image_percentile"	Each image is scaled independently. Looks visually nice but is not directly comparable.	Not applicable.	Use for exploratory visualization only.
CELL_INTENSITY_MAP_STAT = "mean"	Uses average cell intensity. More stable.	Not applicable.	Good default for phenotype maps.
CELL_INTENSITY_MAP_STAT = "p90"	Uses high-intensity cell signal. More sensitive to localized marker expression.	Not applicable.	Use if marker signal is punctate or localized.

Phenotype classification tuning

Parameter	Increase value	Decrease value	Use when
PHENOTYPE_POSITIVE_PERCENTILE	Stricter positivity threshold. Fewer	More permissive positivity threshold.	Adjust when phenotype maps

Parameter	Increase value	Decrease value	Use when
	cells classified as positive.	More cells classified as positive.	look too positive or too negative overall.
PHENOTYPE_POSITIVE_PERCENTILE_BY_CHANNEL[ch]	Makes that specific marker stricter.	Makes that specific marker more permissive.	Use when one marker behaves differently from the others.
PHENOTYPE_THRESHOLD_MODE = "percentile_global"	Uses all cells pooled together to define thresholds. Good for exploratory comparison.	Not applicable.	Good default for testing.
PHENOTYPE_THRESHOLD_MODE = "percentile_per_image"	Each image gets its own threshold. Can hide real differences between images.	Not applicable.	Use only for exploratory visualization.
PHENOTYPE_THRESHOLD_MODE = "manual"	Uses fixed marker thresholds. Best when thresholds are validated.	Not applicable.	Use for final analysis if manual cutoffs are known.
PHENOTYPE_THRESHOLD_MODE = "control"	Uses control images to calculate thresholds. More biologically grounded.	Not applicable.	Use for final biological interpretation when proper controls exist.

Common quick fixes

Problem	Try this first
Colony mask is too small	Decrease THRESH_PCT or try MASK_MODE = "composite"
Colony mask includes background	Increase THRESH_PCT
StarDist under-segments nuclei	Decrease STARDIST_PROB_THRESH; try increasing STARDIST_NMS_THRESH; or use NUC_SEGMENTATION_MODE = "hybrid"
Area filtering removes valid nuclei	Set STARDIST_MIN_SIZE = None and STARDIST_MAX_SIZE = None
Nuclei are merged in watershed fallback	Decrease DENSE_WS_MIN_DISTANCE
Nuclei are over-split in watershed fallback	Increase DENSE_WS_MIN_DISTANCE
Phenotype map is too negative	Lower the marker-specific percentile for

Problem	Try this first
Phenotype map is too positive	that channel Increase the marker-specific percentile for that channel
Cell maps are not comparable across MPs	Use <code>CELL_INTENSITY_NORM_MODE = "global_percentile"</code>
Cell maps look pretty but suspicious	Check whether they are using per-image normalization
Metrics look biologically weird	Check <code>debug/</code> , <code>masks/</code> , <code>qc/</code> , and threshold diagnostics before interpreting metrics

Recommended tuning order

When something looks wrong, tune in this order:

1. Check channel mapping in `debug/`.
2. Fix the colony mask if needed.
3. Fix nuclei segmentation if single-cell outputs are used.
4. Check radial profiles and topography maps.
5. Tune phenotype thresholds only after segmentation looks correct.
6. Interpret metrics only after visual QC passes.